Parameterized models can produce realistic, manipulable images of human faces—with a surprisingly small number of parameters.

Parameterized Models for Facial Animation

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There are several possible approaches to applying computer graphics techniques to facial animation.* One is key frame animation, a technique of conventional animation. The idea behind key frame animation is that the desired image is specified for a certain point in time and then again for another point in time a certain number of frames later. A computer algorithm generates the necessary images for the frames in between these "key frames." The technique has been successfully adapted to computer-assisted two-dimensional animation systems,[1,2] and it has also been successfully applied to three-dimensional facial animation.[3] Its limitation in three dimensions, however, is that each key frame must be completely specified. Developing the required 3-D key frame descriptions is difficult at best; as the number of key frames needed becomes large, the approach becomes impractical.

Another approach is that of using parameterized facial models.[4] These models, which are primarily tools to support computer-assisted animation of realistic, 3-D facial images, have been far more successful than key frame animation. Once an appropriate parameterized model is developed, a wide variety of animation can be produced with relatively little effort.

A "complete" parameterized model for faces would allow the animator to create any facial image by specifying the appropriate set of parameter values. A sequence of images could then be created, modifying the parameters as needed for each image in the desired sequence. Key framing would then be applied to the parameters required to specify the faces rather than directly to the faces.

*Note that *animation* and *simulation* are not the same: The goal of animation is to communicate an idea, story, mood, etc.; the goal of simulation is to exactly model a process, theory, or system. Animation allows and often requires artistic license, i.e., deviation from exact modeling.

Potential applications

In the future, parameterized models might provide previews of the effects of corrective surgical or dental procedures on given faces. Such applications demand biometrically accurate models based on the structures of the hard and soft tissues of the face. Modifications to these structures, of course, affect the observable characteristics of the face. These might be conformation changes or changes in the range of possible expressions. Also, changes in the functionality of facial nerves can affect the expression characteristics of the face. Experiments dealing with the way we perceive faces could also benefit from a model that produces precisely controlled facial images.

Another potential application is data compression for image transmission. Given good parameterized models, parameter extraction algorithms, and sufficient processing power at both the receiving and transmitting ends, it is possible to transmit facial images simply by sending the appropriate parameter values. This would greatly reduce the communication bandwidth required. The success of such an approach requires either a facial model (with a manageable number of parameters) that is nearly complete or the user's acceptance of images that only approximate actual faces; it also implies the existence of techniques to extract the needed parameters from images of actual faces.

Interactive parameter specification with visual feedback and a well-designed human interface would make it possible for an untrained user to create specific facial images. For example, a crime victim could interactively modify parameters to obtain a good 3-D approximation of the face of an assailant. An early 2-D implementation of this idea was developed by Gillenson.[5]

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Parameterized models

There are two basic ideas involved in the creation of parameterized graphical models. The first is the underlying concept of parameterization and the development of appropriate parameter sets. The second is that of graphic image synthesizers that produce images based on some defined parameterization.

The basic parameterization concept. Consider a class of objects in which individual members have identifiable differences. If the differences between member objects can be associated with a set of differentiation or specification criteria, individual members of the class can be described or specified by the criteria values. These criteria values can be thought of and referred to as parameter values.

A "complete" set of criteria or parameters is one that allows every member of an object class to be specified by selecting appropriate parameter values. If certain members of the class can be differentiated but not uniquely described by parameter values, the parameter set is not complete. In a complete set, every possible unique member of the class can be described by a unique n-tuple of parameter values. Some parameters might have values from finite discrete sets, while others might have values from bounded or unbounded continuous ranges.

A simple example of this notion is the class of objects called cubes. The description or specification for a particular cube consists of a set of such parameter values as the length of an edge, the color of each face, the weight, material, etc. If the set of parameters is complete, one could specify every possible cube by specifying the appropriate n-tuple of the parameter values. However, even for such simple objects, it can be difficult to develop truly complete parameterizations.

The image synthesis model. Given a parameterization scheme, the next task is to develop a synthesis model to produce images based on the parameter values. Figure 1 illustrates the major components of animator interaction with such a model. It has two major parts, the parametric model and the graphics routines.

The heart of the process is the set of algorithms, functions, and data that form the parametric model. This model takes as input the parametric values specified by the animator. It produces as output a set of graphic primitives, such as vector and polygon descriptors. These descriptors are passed as input to the graphics routines that actually create the images. In an interactive environment, the animator can adjust the parameters as expressed in the generated images. In animation models, the model designer and the animator interact closely in determining the actual characteristics of the model.

Part of the model development process is determining the type of images to be produced. These might be simple vector images generated from line segments, or, on the other end of the complexity spectrum, they might be shaded images based on quadric or cubic surfaces with texture rendered through complex lighting models.[6] A typical choice is polygonal surfaces rendered with fairly simple shading rules.

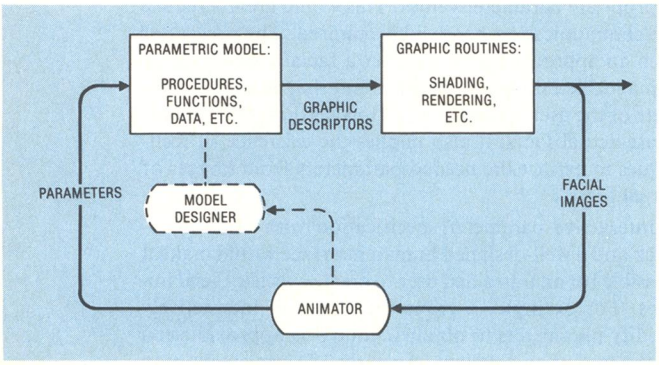


Figure 1. Parameterized image synthesis model for facial animation.

Facial parameterization

Given the conceptual framework outlined above, developing the desired parameterized models consists of two distinct tasks: developing appropriate parameters and developing image synthesis models based on these parameters.

The first step is to determine the appropriate set of facial parameters—a nontrivial task. Ideally, one would develop a complete parameter set for creating and specifying any possible face with any possible expression. The possiblility of developing such a set is an open question.

If this ideal is not possible or even practical, can useful parameterized models be developed? Furthermore, can useful parameter sets be developed in which the number of parameters is relatively small? For facial animation, useful, incomplete sets with only a few parameters have been demonstrated.[4,7]

Research and development aimed at improving the quality and scope of these models continues. By *quality,* we mean the appropriateness and efficiency of the parameters. That is, are they the right ones, are they natural to use, do they meet immediate needs? *Scope* refers to the range of faces and expressions that can be generated under the chosen parameters.

Developing parameter sets. How is a facial parameter set developed? One approach is to simply observe the surface properties of faces and develop ad hoc sets that allow these observed characteristics to be specified parametrically. This approach is unlikely to produce complete sets, however. A second approach involves studying the underlying structure, or facial anatomy, and developing a set of parameters based on it.

The models developed by Platt and Badler[8], for example, deal directly with the underlying structures that cause facial expression. Their work uses a notational system to encode the actions performed by the face. The notation drives a model of the underlying muscle structure, which in turn determines the facial expression.

A third approach to developing a facial parameter set blends these first two. Parameters are based on structural understanding, wherever possible, and are supplemented

as necessary by parameters based on observation. The models we have developed to date fall into this third category.

There are two broad categories of parameters: those controlling the *conformation*, or structure, of an individual face, and those controlling its *expression*, or emotional content. To a certain extent, these two categories overlap, but conceptually they can be considered distinct.

The most general parameterized models are those allowing wide ranges of both facial conformation and expression. However, models that allow variation only in expression or only in conformation can be sufficient for certain applications. The animation of a specific character might require variation only in facial expression, for example. The advantages of limited models include simplicity and efficiency.

What should the parameters for an expression model be? In general, this is an open question, although the structure-based Facial Action Coding System notation developed by Ekman is a possible foundation for such a model.[9,10] In the FACS system there are relatively few—about 50—independent facial actions. These can occur separately or in combinations to form expressions. The actions are based on the activity of facial muscles, and many of them normally take place only in combination. With training, however, it is possible to control each of these actions independently.

The FACS system appears to be the best basis, at present, for complete expression models. Direct parameterized models based on this system have not been developed to date, although the Platt and Badler structure-based facial model does use FACS notation.

Expression parameters. Which parameters are best included in a simple expression model remains unresolved. The parameters that have been used in past and current models are discussed below. Of primary importance in expression are the eyes and mouth,[11,12] so most expression parameters relate to these areas.

Expression parameters found useful for the eyes include pupil dilation, eyelid opening, eyebrow position and shape, and the direction in which the eyes are looking.

Useful expression parameters for the mouth include jaw rotation (which controls mouth opening), width of the mouth, mouth expression (smiling, frowning, etc.), position of the upper lip, and positions of the corners of the mouth.

Other useful expression parameters include the size of the nostrils and the orientation of the face (head) with respect to the neck and the rest of the body. The ability to orient the face—to turn and tilt it—was not included in early models and was obvious by its absence. With 15 or so parameters, it is possible to develop a model that allows interesting expression animation and can be animated to a spoken soundtrack.[7]

Conformation parameters. Changes in the conformation of faces (those aspects that vary from individual to individual and make each person unique) require a different set of parameters. Again, the ideal set is unknown. The following parameters continue to be used in current models, and although this parameterization is clearly not complete, it does allow for a wide variety of facial conformation within the implied limits.

Some conformation parameters apply globally to the face. In addition to skin color and the aspect ratio of the face (height to width), these global parameters include a transformation, suggested by other researchers,[13] that attempts to model facial growth. Conformation parameters control the color (and the texture in more elaborate models) of the eyebrows, eyelashes, iris, lips, and other features.

Other conformation parameters use relative size (scale), shape, and positioning information to control

- neck length and shape;
- chin, forehead, cheek, and cheekbone shape;
- eyelid, eyeball, and iris size and the position and separation of the eyes;
- jaw width;
- nose length and the width of the bridge and end of the nose; and
- chin and forehead scale and the scale of the mouth-to-eyes portion of the face *vis-à-vis* the rest of the face.

The development of truly complete conformation parameter sets appears very difficult. Little in the way of theory exists to support their development, and the variations in facial structure from one individual to another are far less understood than the ways in which a given structure varies from one expression to another.

Viewer expectations. Sensitive to subtle variations in expression and conformation, we all continually observe faces and develop very clear expectations about them. Facial expression is an important communication channel; in some contexts, it takes priority over other channels (words, for example).

An interesting and slightly frustrating phenomenon we observed while developing our models suggests the following rule: The closer the images get to reality, the more critical the viewer becomes. If the images are clearly perceived as artificial or synthetic, the viewer seems willing to be somewhat forgiving and accept them as such. If the image is clearly a caricature, the viewer recognizes and accepts this. But if the image is clearly supposed to be realistic, the viewer is very sensitive to any flaws.

A model for facial image synthesis

Formal procedures for developing a model that can generate facial images based on a parameter set do not currently exist. We have used the model described below to generate the images shown in Figures 3 through 10. Based on a parameter set that includes both expression and conformation parameters similar to those discussed above, this model, for a specific character, was derived from earlier, more general versions. Even though the current model is used primarily for expression and speech animation, it retains many of the conformation parameters of its predecessors.

Polygon topology. For this model, image generation is based on surfaces constructed from connected networks of polygons. The facial mask is one polygonal surface, and each eyeball is another; teeth are separate polygons. The polygonal connection, or topology, of each surface depicted in Figure 2 remains constant, but the three-dimensional position of each polygon vertex varies according to the parameter values, eye orientation, and face orientation. As the vertex positions change, the polygonal surfaces flex and stretch to allow the face to change.

As shown in Figure 2, the polygon topology for this model is an arbitrary network rather than a regular grid. The polygons are sized and positioned to match the features of the face. There are many small polygons in areas of high curvature and fewer, larger polygons in the flatter regions. Also, polygon boundaries are designed to coincide with the desired creases in the face. This topology is the result of trial-and-error experience gained with earlier models.

In our earlier models for facial image synthesis, the vertex position values were determined by photogrammetrically measuring the surfaces of real faces.[3,4] For the character model displayed in Figures 3 through 10, the vertex positions were interactively adjusted to match sketches supplied by the character design artist.

This model is implemented as a C program that is both data and parameter driven. Data files define the polygon topology and the extreme vertex positions used for interpolation. The parameter values can be entered interactively or, for animated sequences, from command files. The program generates polygon descriptors, which drive polygon rendering routines.

Operations. Five types of operations determine vertex positions from the parameter values. Although sometimes applied globally to the face, instances of these operations are in most cases applied independently to specific facial regions.

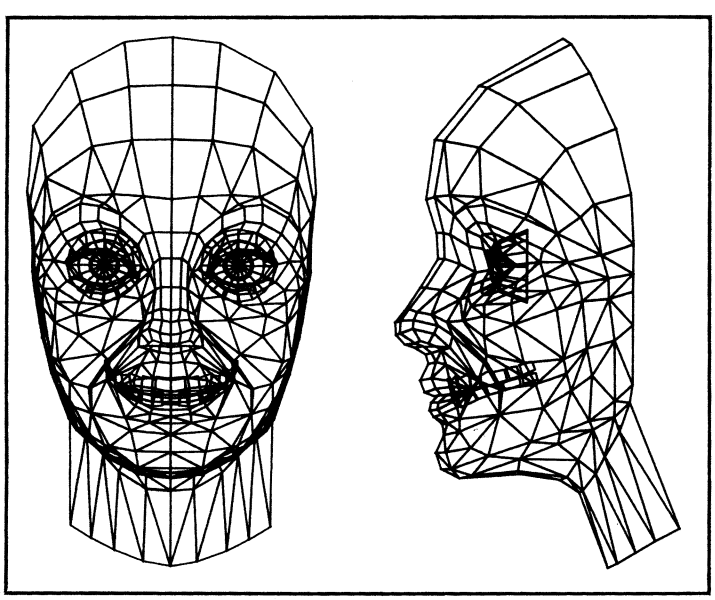


Figure 2. Front and side views of the polygonal network, or topology, used to generate the images shown in Figures 3 through 10.

(1) *Procedural construction* is used to model the eyes. The eyeball procedure accepts parameters for eyeball, iris, and pupil size and for the color of the iris, the eye position, and eyeball orientation. It then procedurally generates the polygon descriptors needed to create the desired eyes.

(2) *Interpolation* is used for those portions of the face that change shape. These include the forehead, the cheekbone region, the neck, and the mouth. Each of these areas is independently interpolated between defined extreme shapes and associated with a parameter value. For a given vertex in one of these regions, two positions are defined—one for each extreme. The parameter value controls interpolation between the extreme positions.

(3) *Rotation* is used to open the mouth by rotating the lower portion of the face around a jaw pivot axis.

(4) *Scaling* controls the relative size of facial features—the chin's prominence and the widths of the nose, jaw, and mouth. The scale factors are applied only to vertices within the specified regions of the face.

(5) *Position offset* controls the length of the nose, the corners of the mouth, and the raising of the upper lip. Position offsets move collections or regions of vertices as a group.

Model faces. The rendering technique for Figures 3 through 10 is Phong shading,[14] which allows surface highlights. The position of the light source and overall orientation of the face are parameters external to the face model.

Figure 3 shows the face in its neutral expression, with the eyes closed. In Figure 4, the eyes are almost fully open and the face is rendered in faceted shading to show the polygon structure.

The eyelids are unusual in that they are implemented by a combination of interpolation and procedural techniques and are defined in only two dimensions. As controlled by the eyelid parameter value, the eyelids are interpolated between two extremes, open and closed, and the resulting two-dimensional vertex positions are then procedurally projected onto spherical surfaces centered on and slightly larger than the eyeballs.

The faces shown in Figures 3 and 5 differ only in their values for two parameters—the eyelid-opening and mouth expression parameters. Mouth expression is interpolated between two extremes, which in this case are a slight smile and a frown.

Different values for six parameters distinguish Figure 6 from Figure 3. In Figure 6, the eyes are wide open, the eyebrows are raised, the jaw is rotated to open the mouth, the upper lip is raised, the mouth is narrowed, and the pupils are dilated. The eyebrows are manipulated by interpolation, the upper lip is raised via positional offsets, and the mouth is narrowed by scaling.

The mouth is opened by rotating the vertices of the lower part of the face about a jaw pivot axis. This rotation is tapered. The vertices along the jaw are rotated by the full amount specified in the rotation parameter, but the

Figure 3. Face in its neutral expression with the eyelids closed.

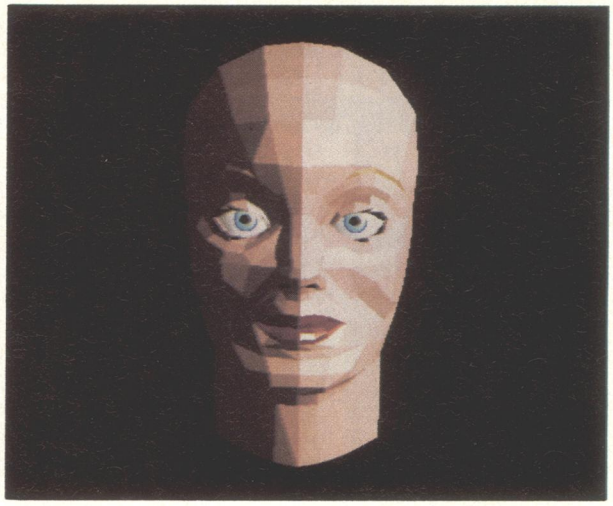


Figure 4. Face with the eyelids open, rendered in faceted shading.

Figure 5. A frowning face.

Figure 6. Face with the eyelids wide open, eyebrows raised, narrow open mouth, raised upper lip, and dilated pupils.

Figure 7. A younger face.

Figure 8. Face with a shorter nose.

Figure 9. Face with a smaller chin.

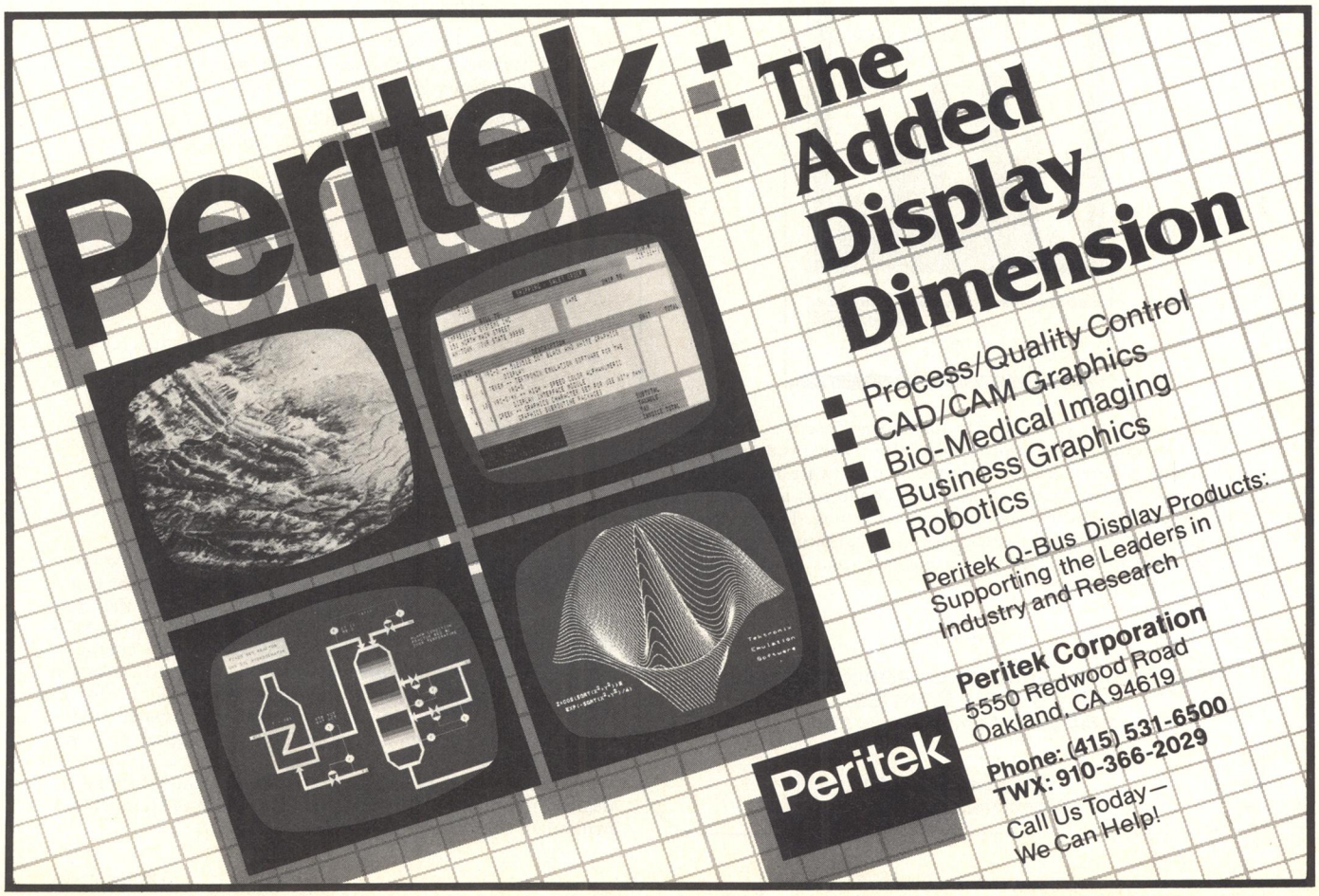
Figure 10. Modified facial conformation with the same expression as Figure 6.

rotation applied to those points above the jaw and below the mouth *tapers* so that the amount of rotation gradually diminishes for points higher in the face. The lower lip rotates, the upper lip does not; therefore, the mouth opens.

Figures 7 through 10 show how several conformation parameters can modify the face. Figure 7 results from modifying the parameter that controls a global facial growth transformation.[13] Applied to the entire face, this transformation approximates structural growth from young child to young adult. The intent in Figure 7 is to show a somewhat younger face.

In Figure 8, the length of the nose was shortened by adjusting a single offset parameter. In Figure 9, the size of the chin was modified by adjusting a single scaling parameter value.

In Figure 10, the expression parameters are identical to those of Figure 6. However, five conformation param-



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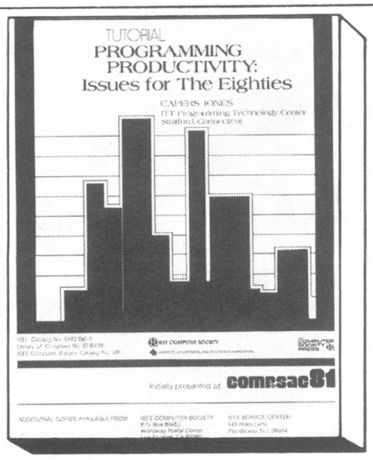
eters have been modified: the length of the nose, the width of the jaw, the size of the chin, the height of the forehead, and the distance between the mouth and the eyes. Scale factors determine jaw width and mouth-to-eye distance; interpolation controls the height of the forehead.

Parameterized models are powerful tools for facial image synthesis and animation, even as implemented in the relatively primitive models developed to date. The value of parameterized models to the animator is that he or she need only manipulate a limited amount of information—the parameters—to create a sequence of images. With the development of better, more complete facial parameter sets and the application of more sophisticated image synthesis techniques, parameterized models will allow even better facial animation. ■

Acknowledgments

The images shown in Figures 3 through 10 were produced with polygon rendering programs developed by Paul Heckbert at the New York Institute of Technology Computer Graphics Lab. Bil Maher, also of the NYIT Computer Graphics Lab, created the character design.



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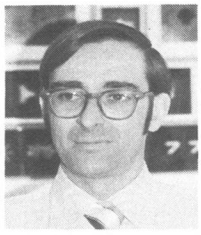

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